

Tiffany Sun

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EDUCATION

Amherst College, Amherst, MA

Expected May 2028

- Bachelor of Arts | Math and Computer Science Major | GPA: 3.60
- **Relevant Coursework:** Probability, Data Science, Linear Algebra, Intermediate Statistics, Multivariable Calculus, Introduction to Computer Science (I & II), Evolutionary Computation
- **Activities and Clubs:** Amherst College Emergency Medical Services (ACEMS), Women in Computer Science (WiCS)

CodePath

June 2026 - August 2026

- **Relevant Coursework:** Applications of AI Engineering, Introduction to Web Development

RELEVANT EXPERIENCE

Amherst College, Mathematics and Statistics Department, Amherst, MA

September 2025 - May 2026

Statistics and Data Science Fellow

- Built logistic and quantile regression models to analyze donor behavior and identify key predictors of giving patterns (e.g., engagement, prospect status, ask amounts); presented findings to a board of stakeholders at the end of the semester.
- Developed reproducible data cleaning and preprocessing pipelines; collaborated via Git to improve data quality and reliability across multiple data sources.

Amherst College, Wellbeing and Emotion Change Lab, Amherst, MA

June 2025 - August 2025

Summer Undergraduate Research Fellow

- Designed and executed an independent research project analyzing emotional beliefs using correlational analyses; presented findings to 100+ attendees at Amherst's annual research symposium. Qualitatively coded 500+ survey responses to support longitudinal analyses; improved data completeness across participant cohorts through targeted outreach.

Amherst College, Mathematics and Statistics Department, Amherst, MA

September 2024 - January 2025

Gregory S. Call Academic Intern

- Built and evaluated random forest models in R (500 decision trees, ~754-record dataset) to predict live birth and clinical pregnancy outcomes, achieving ~90+% accuracy.

PROJECTS

Starbucks Customer Analytics - Python, Pandas, NumPy, Seaborn

July 2026

- Cleaned and transformed a 100,000-order Starbucks transaction dataset from Kaggle into recency, frequency, and monetary features per customer; built a quantile-based RFM scoring system to segment 15,000 customers, identifying 38% at churn risk.
- Published end-to-end analysis on Medium and GitHub, documenting the full pipeline from data cleaning through RFM segmentation and churn insights.

NYC CrashLens - R, Plotly, Leaflet, Shiny R

June 2026 - July 2026

- Processed and wrangled 2.27M NYPD-reported crash records (2012–2026) to build a full-stack interactive analytics platform in R Shiny. Features include a choropleth map dashboard, regional comparator, and filterable database with CSV export, allowing users to explore casualty, causal, and spatiotemporal trends across NYC.

Predicting Heart Disease - R

December 2024

Final Project for Intermediate Statistics (STATS-230)

- Applied stepwise AIC selection, VIF testing, and likelihood ratio tests to build an optimized logistic regression model predicting heart disease risk; identified a significant interaction between gender and resting blood pressure and translated findings into clinically meaningful insights.

CERTIFICATIONS

IBM SkillsBuild Data Analytics Certificate

June 2026 - July 2026

SKILLS

- *Programming & Data:* Python, R, SQL, Java, HTML, CSS, JavaScript
- *Tools & Platforms:* Git/GitHub, Excel, PowerPoint, Google Workspace, Notion, Canva, Figma, SPSS, Qualtrics, AI Tools
- *Language Skills:* Mandarin (Fluent), German (Beginner)